# Block diagram

*sheet symbols open the pages; arrows are the interfaces between them*

## Power and ESP32-S3 DevKit

ESP32-S3 DevKitC-1 (N16R8): supply pins, reserved GPIO

5V0 -> FB1 -> LP5907 -> 3V3_RADIO -> VMCU_IN

DevKit 3V3 decoupling, test points

Fájl: power.kicad_sch

*The ESP32 module is split in four units, each on the page it serves:*
*IC1A supply (p2), IC1B radio link (p3), IC1C user interface (p4), IC1D reserved (p2).*

3V3_RADIO, VMCU_IN

5V0, 3V3

## Radio board (BRD4265B)

P200/P201 to the BRD4265B (EFR32FG23)

FG23 SPI (IO_MUX slave) + UART + RDY

SWD bit-bang, I2C board-ID, cut points JP1-JP8

Fájl: radio.kicad_sch

I2C_SDA/SCL (OLED shares the bus)

## User interface

ER-TFTM024-3 TFT, 4-wire SPI + XPT2046 touch

SSD1306 OLED on I2C 0x3C

two ADC1 button ladders (6 keys)

Fájl: ui.kicad_sch

# Mechanical

*M3 mounting holes, not plated to GND*

H1 M3 H2 M3 H3 M3 H4 M3

# Design notes

*Net names are global (labels and rail symbols): layout, twin and firmware share one name per net.*

*A label sits on a short stub at a pin end; wires stay inside a block; supplies only as rail symbols.*

*GPIO plan: FG23 link on the SPI2 IO_MUX pins 10/11/12 + UART TX 13 + RDY 14; TFT on SPI3 via the matrix (4-7, 15-18); touch 8/9/21; buttons ADC1 1/2; I2C 41/42; SWD 38/39/48, RESETn 40.*

*Forbidden: strapping 0/3/45/46, octal PSRAM 35-37 (N16R8). Reserved: UART0 43/44, USB 19/20, WS2812 47.*

*Cut points JP1-JP8 isolate every radio-board signal; R1-R5 are 0R links on a damping-resistor footprint.*

*Assembly: R26/R27 DNP, value NA(4.7k). Values carry the critical parameter (value/voltage/tolerance).*

*FB1: value 600R@100MHz but MPN field BLM18SP300SH1D (30R) - resolve before ordering.*

ESP32-S3 DevKitC-1 carrier for the Silabs BRD4265B (EFR32FG23) radio board

**RF Sparkling Kft. - HA7DCD**

Sheet: /

File: ionosSDR_mini.kicad_sch

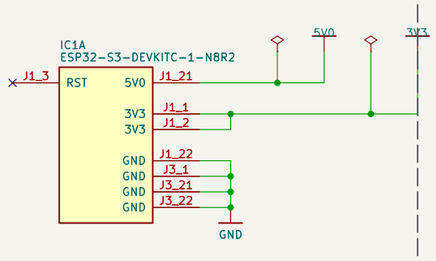
***Title: ionos-sdr-mini rev-B3 - Overview***

| Size: A3 | Date: 2026-10-09 | **Rev: B3** |
|---|---|---|
| KiCad E.D.A. 10.0.6 | | Id: 1/4 |

## ESP32-S3 DevKitC-1

*supply pins: the DevKit is the power source*

*RST = EN, button on the DevKit*

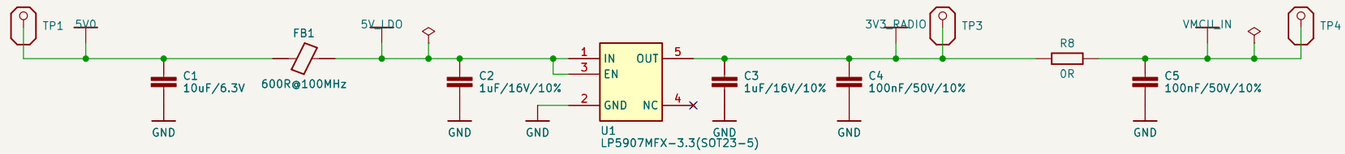



*GND test point and ERC flag*




## Radio supply

*5V0 -> FB1 -> LP5907 -> 3V3_RADIO -> R8 -> VMCU_IN*



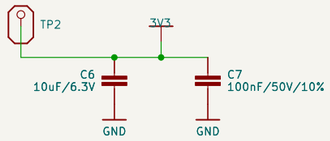

*FB1 + C1/C2 keep DevKit 5 V noise out of the LDO*

*LP5907: 250 mA, 6.5 uVrms, EN tied to IN*

*R8 0R fitted: VMCU_IN = 3V3_RADIO (P201 pin 2)*

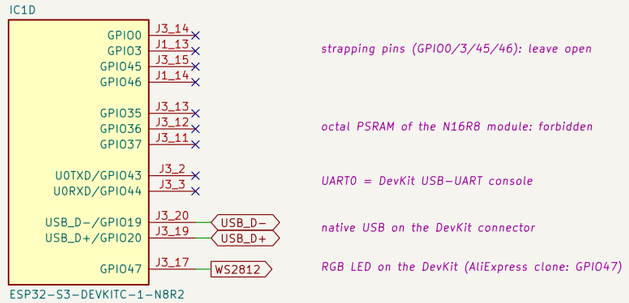
## 3V3 decoupling

*DevKit LDO output; displays and pull-ups*




## Reserved GPIO

*not used on this board — keep it that way*




*strapping pins (GPIO0/3/45/46): leave open*

*octal PSRAM of the N16R8 module: forbidden*

*UART0 = DevKit USB-UART console*

*native USB on the DevKit connector*

*RGB LED on the DevKit (AliExpress clone: GPIO47)*

DevKit 5V0/3V3 in, 5 V filter, LP5907 radio LDO, reserved GPIO

**RF Sparkling Kft. – HA7DCD**

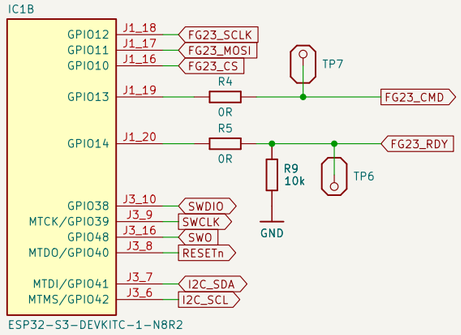
Sheet: /Power and ESP32-S3 DevKit/
File: power.kicad_sch

**Title: ionos-sdr-mini rev-B3 – Power and ESP32-S3 DevKit**

| Size: A3 | Date: 2026-10-09 | **Rev: B3** |
|---|---|---|
| KiCad E.D.A. 10.0.6 | | Id: 2/4 |

## ESP32-S3 radio link

*SPI2 IO_MUX slave, UART, SWD, I2C*




*ESP32 is the SPI slave (FG23 = master).*
*IO_MUX pins only — no GPIO matrix.*

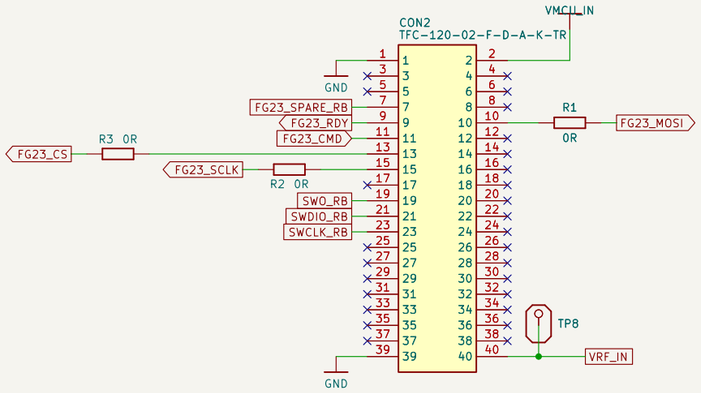
*R4/R5 near the ESP32; R9 near P201.*

*SWD is bit-banged; RESETn open-drain.*

*Laid out for the AliExpress DevKit clone (LED on GPIO47). Official DevKitC-1 puts the WS2812 on GPIO38 (v1.1) / GPIO48 (v1.0) = SWDIO / SWO here: remove that LED or use the clone.*

## P201 — radio board header

*FG23 link, SWD, VMCU (BRD4265B expansion)*



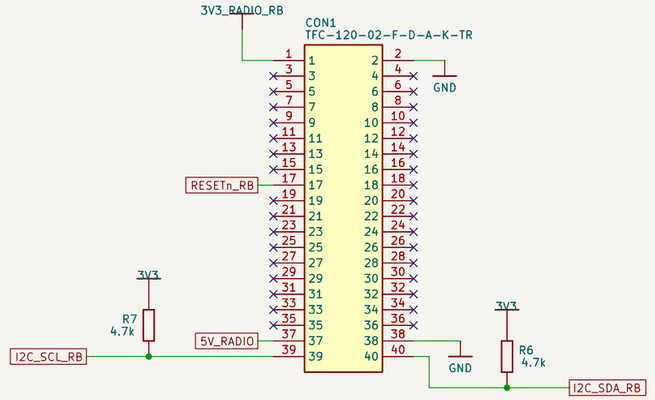

*R1–R3 sit at P201: 0R fitted,*
*22–68R damping footprint (2-layer build: 68R).*

*Pinout = Silabs BRD4265B / WSTK expansion header.*

*VRF_IN (P201.40): track to TP8 only — a test point on the BRD4265B, not connected there.*

## P200 — radio board header

*supply, reset, I2C (board-ID EEPROM)*





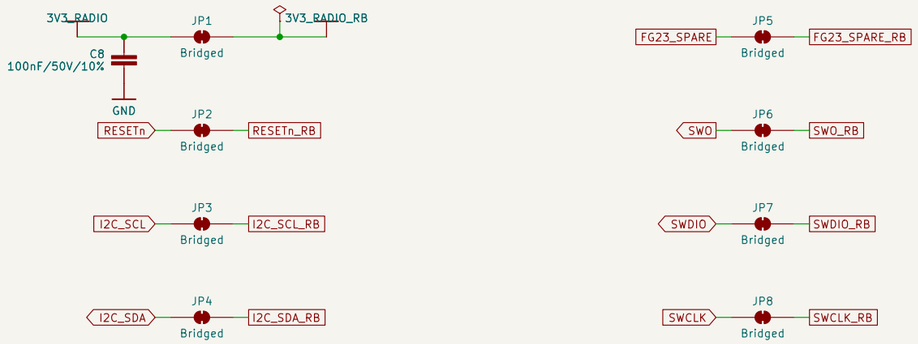


*R33 not fitted: DevKit 5 V to P200.37, unused on the BRD4265B.*

*R6/R7: I2C pull-ups on the radio side. Board-ID EEPROM on the radio board, OLED 0x3C on ours.*

## Cut points

*one solder jumper per radio-board signal, bridged by default*

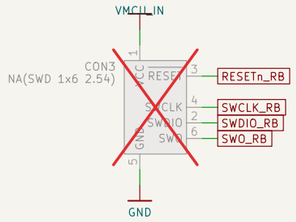



*FG23_SPARE: track and test pad only, no part fitted.*

## SWD debug header

*external debugger; radio side of the cut points*

*1x6 2.54 mm row, pin order of the Tag-Connect TC2030-CTX:*
*1 VTref (VMCU)  2 SWDIO  3 nRESET  4 SWCLK  5 GND  6 SWO*




*Footprint only (NA): pogo clip or solder-in pins.*

*With a debugger on CON3 the ESP32 SWD pins must stay inputs (firmware), or cut JP2/JP6–JP8.*

*VTref senses VMCU; the debugger must not power it.*

P200/P201 mezzanine, FG23 link, SWD, cut points
**RF Sparkling Kft. — HA7DCD**

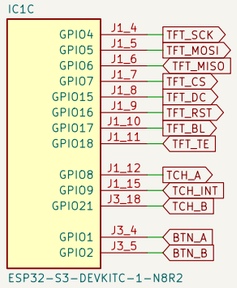
Sheet: /Radio board (BRD4265B)/
File: radio.kicad_sch

***Title: ionos-sdr-mini rev-B3 — Radio board (BRD4265B)***

| Size: A3 | Date: 2026-10-09 | **Rev: B3** |
|---|---|---|
| KiCad E.D.A. 10.0.6 | | Id: 3/4 |

## ESP32-S3 user interface

*TFT on SPI3 via the GPIO matrix*

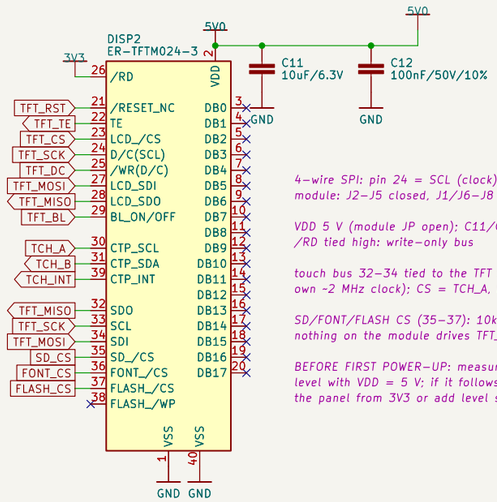



*Touch, no jumper — firmware picks the mode:*
*TCH_A GPIO8 resistive CS / CTP SCL*
*TCH_B GPIO21 PENIRQ / CTP SDA*
*TCH_INT GPIO9 unused / CTP INT*
*R31/R32 4.7k fitted in every build.*

*BTN_A/BTN_B: ADC1, usable with WiFi on.*

## TFT ER-TFTM024-3

*2.4 in 240x320, 4-wire SPI, resistive touch (XPT2046)*




*4-wire SPI: pin 24 = SCL (clock), 25 = D/C*
*module: J2–J5 closed, J1/J6–J8 open*

*VDD 5 V (module JP open): C11/C12 at the header*
*/RD tied high: write-only bus*

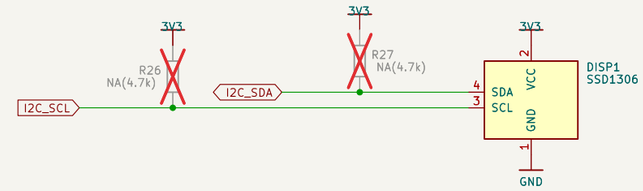
*touch bus 32–34 tied to the TFT SPI (XPT2046,*
*own ~2 MHz clock): CS = TCH_A, PENIRQ = TCH_B*

*SD/FONT/FLASH CS (35–37): 10k pull-ups, so*
*nothing on the module drives TFT_MISO*

*BEFORE FIRST POWER-UP: measure MISO/TE high*
*level with VDD = 5 V; if it follows 5 V, run*
*the panel from 3V3 or add level shifting.*

## OLED SSD1306

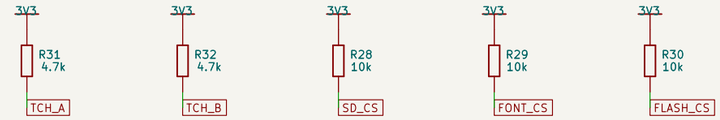
*0.96 in 128x64, I2C 0x3C*




*R26/R27 not fitted: the bus pull-ups are R6/R7*
*on the radio side; fit them if JP3/JP4 are cut.*

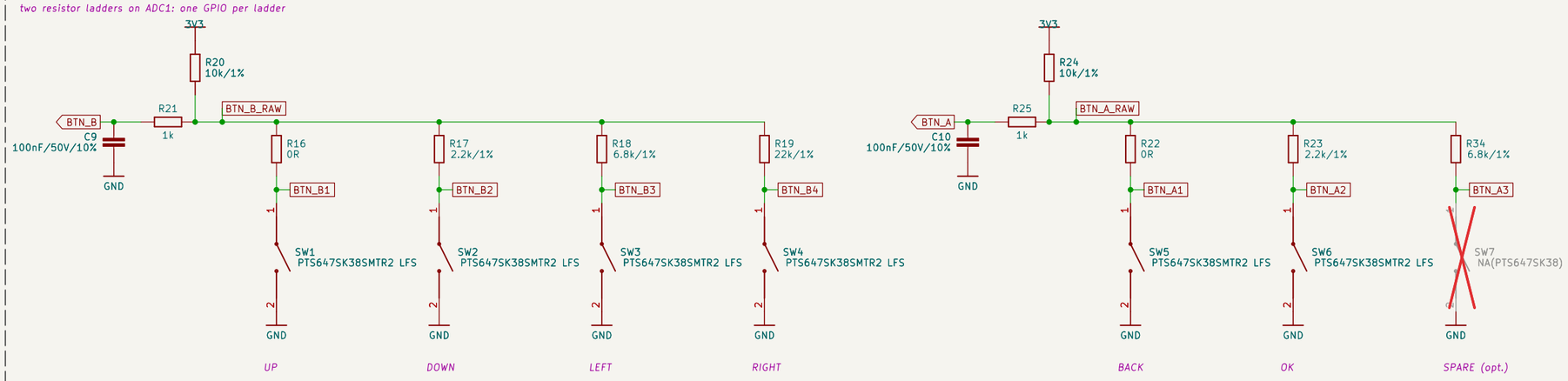
## Header pull-ups

*touch CS/IRQ and the module flash/font/SD chip selects*




## Buttons

*two resistor ladders on ADC1: one GPIO per ladder*




*Both ladders: 0R / 2.2k / 6.8k / 22k under a 10k pull-up -> one firmware table;*
*levels 0 / 0.60 / 1.34 / 2.27 / 3.3 V, thresholds 0.30 / 0.97 / 1.80 / 2.79 V. Wake: 0R keys only.*

TFT + resistive touch, OLED, two ADC button ladders

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Sheet: /User interface/
File: ui.kicad_sch

***Title: ionos-sdr-mini rev-B3 — User interface***

| Size: A3 | Date: 2026-10-09 | **Rev: B3** |
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